

Typography is a Cognitive Tool, Not an Aesthetic Choice

An evidence-based framework
for accessible digital design and
reduced cognitive load.

Synthesising empirical data from Hojjati & Muniandy
(2014) and Rello & Baeza-Yates (2016).

Dyslexia - A widespread hidden disability

Dyslexia is a common, lifelong neurodevelopmental learning disorder that causes significant difficulty with reading, spelling, and decoding words due to challenges processing sounds, not low intelligence.

10% of the
population
have dyslexia

70% of learning
disabilities are
dyslexia



The digital reading penalty

The Print Legacy-

Printed documents utilise an average resolution of 180 to 300 dpi.

The eye traverses the page with minimal hinderance and creates a baseline for human reading speed.

The Screen Reality-

The screen text fundamentally changes the visuals. Due to backlighting pixeation and other factors. Reading speeds decreases by 30%.

↓ -30%



Print Legacy vs Digital Native architecture

Times New Roman (Serif)



Design Intent: Craeted specifically for print media and newspaper

Architecture: Designed to maintain legibilty whil;e cramming high density text into limited space

Digital Reality: Touching letters and serifs create crowding on digital screens

Verdana (Sans Serif)



Design Intent: Engineered exclusively for computer screen displays

Architecture: Features a massively enlarged x height, extensive letter spacing and characters that do ot touch.

Digital Reality: Bolds are enhanced and lowercase letter are instantly distinguishable from uppcase.

Design study - 96 participants , 12 fonts

The first experiment using eye-tracking to measure typeface impact on screen reading for people with dyslexia. Participants read 12 comparable 60-word texts while the Tobii 1750 recorded eye movements.

Group D — Dyslexia

48 participants (22F, 26M), ages 11–50 (mean 20.96). Confirmed clinical diagnosis.

Group D (48 participants)	22 female, 26 male Age: range from 11 to 50 ($\bar{x} = 20.96$, $s = 9.98$) Education: high school (26), university (19), no higher education (3)
Group N (49 participants)	28 female, 21 male Age: range from 11 to 54 ($\bar{x} = 29.20$, $s = 9.03$) Education: high school (17), university (27), no higher education (5)

Group N — Control

49 participants (28F, 21M), ages 11–54 (mean 29.20). No dyslexia diagnosis.

The 12 fonts tested-

Aa

Arial- sans serif
most common

Aa

Helvetica- Apple
design standard

Aa

Courier-
monospaced serif

Aa

Verdana-
recommended font

Aa

Times serif- serif
most common

Aa

Open Dyslexic-
made for dyslexia

Aa

Arial Italic- italic
version of original

Aa

Times italic- italic
version of original

Aa

**Open Dyslexic
italic**- italic

Aa

CMU- computer
modern unicode

Aa

Garamond- old
style serif typeface

Aa

Myriad- humanist
sans serif

Key Findings

Participants with dyslexia had significantly longer reading times overall (mean 24.94s) vs. controls (17.69s, $p < 0.001$). Spearman correlation between groups: $\rho = 0.657$ ($p = 0.024$)

Table II. Median, Mean, and Standard Deviation of *Reading Time* in Seconds for Groups N and D

<i>Font Type</i> Group D				<i>Font Type</i> Group N			
	<i>Median</i>	<i>Mean $\pm$ SD</i>	<i>%</i>		<i>Median</i>	<i>Mean $\pm$ SD</i>	<i>%</i>
	<i>Reading Time</i>				<i>Reading Time</i>		
Arial	24.22	28.35 $\pm$ 12.39	100	Arial	15.63	11.83 $\pm$ 4.32	100
OpenDys	23.81	29.17 $\pm$ 15.79	103	Helvetica	16.78	12.41 $\pm$ 4.17	105
CMU	26.06	29.58 $\pm$ 12.05	104	CMU	17.21	13.03 $\pm$ 4.29	110
Courier	29.73	29.61 $\pm$ 10.87	104	OpenDys	16.34	13.09 $\pm$ 5.85	111
OpenDys It.	25.44	29.68 $\pm$ 14.44	105	Garamond	16.71	13.33 $\pm$ 4.38	113
Helvetica	27.18	31.05 $\pm$ 15.04	109	Courier	16.90	13.75 $\pm$ 5.50	116
Verdana	28.97	31.16 $\pm$ 13.03	110	Myriad	17.86	13.87 $\pm$ 5.32	117
Times	29.30	31.68 $\pm$ 11.81	112	OpenDys It.	17.63	14.34 $\pm$ 6.14	121
Times It.	28.55	32.38 $\pm$ 12.34	114	Times It.	18.50	14.55 $\pm$ 6.38	123
Myriad	26.95	32.66 $\pm$ 14.80	115	Times	18.35	14.75 $\pm$ 5.09	125
Garamond	30.53	33.30 $\pm$ 15.45	117	Verdana	18.86	15.69 $\pm$ 7.88	133
Arial It.	29.68	34.99 $\pm$ 16.60	123	Arial It.	21.25	17.24 $\pm$ 7.53	146

Graph Analysis

- Arial Italic had the longest reading time — significantly worse than Arial ($p = 0.011$), CMU ($p = 0.011$), and Helvetica ($p = 0.034$) for Group D
- Group N: Arial Italic significantly worse than 6 other fonts
- The most legible font however was Arial with the shortest mean reading time

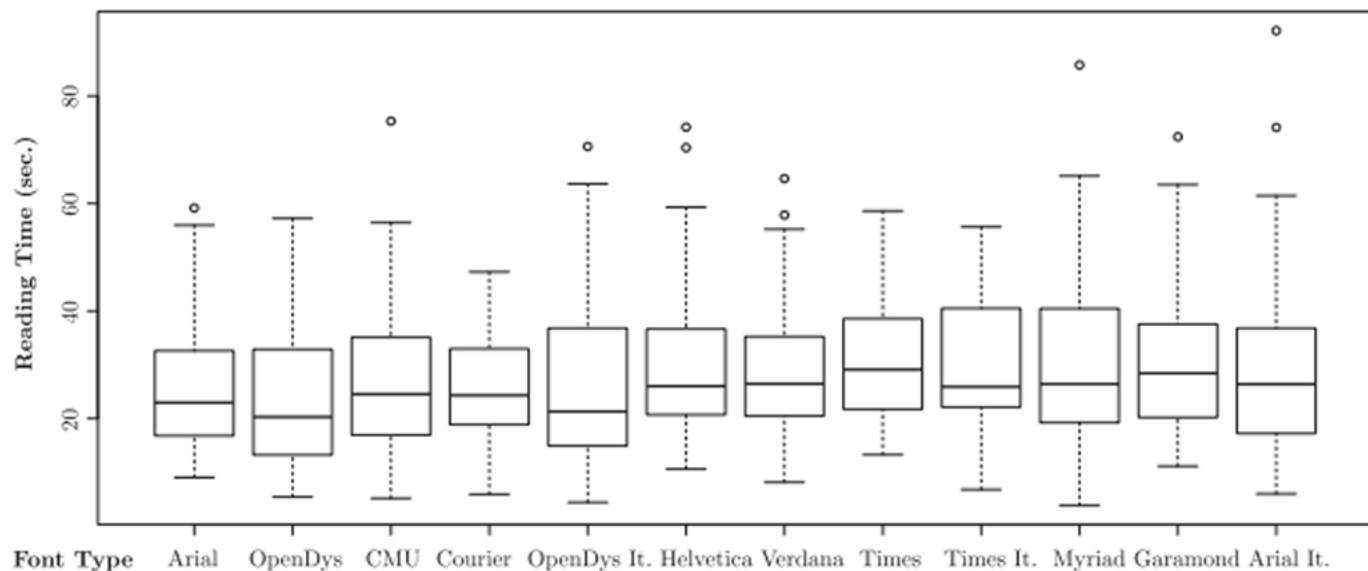


Fig. 4. *Reading Time* box plots by *Font Type* for group D (ordered by average *Reading Time* for group D).

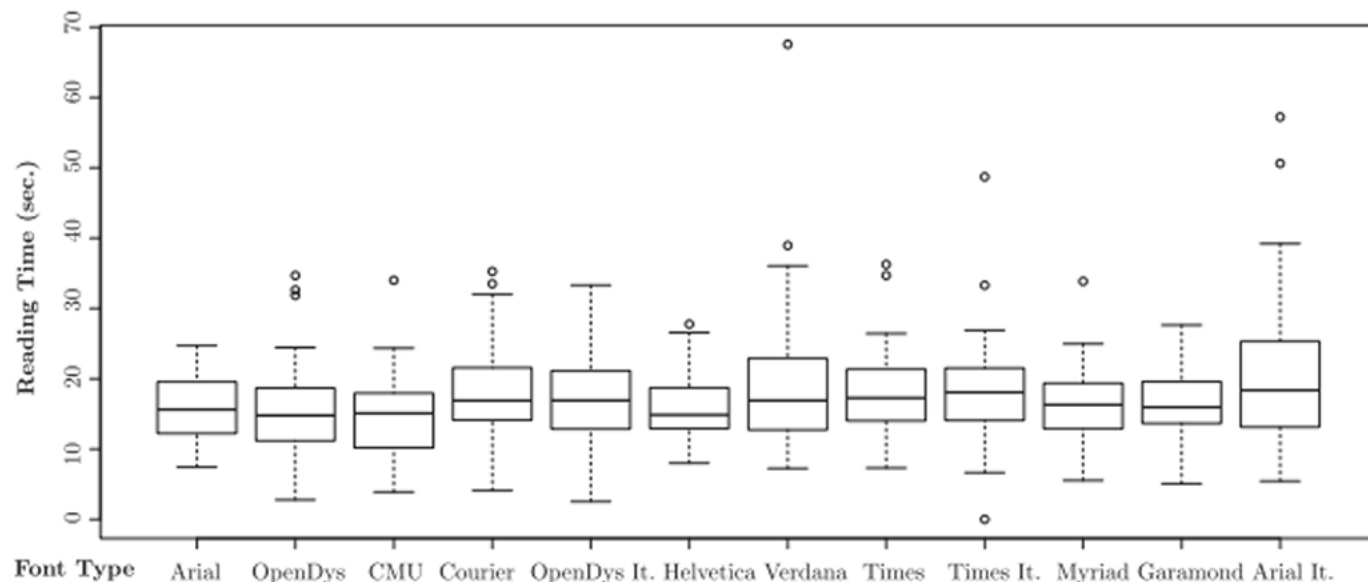


Fig. 5. *Reading Time* box plots by *Font Type* for group N (ordered by average *Reading Time* for group D).

What Number of Fixation Means

The number of fixations is an eye-tracking metric measuring how many times a user's gaze pauses on a specific object

CMU and Arial: had the least fixations for both groups respectively, along with OpenDys coming in a close second for group D backing up the fonts true purpose.

Arial Italic: Being consistently in the bottom half of the groups, however Verdana scored lowest in both groups consistently.

<i>Font Type</i> Group D				<i>Font Type</i> Group N			
	<i>Median</i>	<i>Mean ± SD</i>	<i>%</i>		<i>Median</i>	<i>Mean ± SD</i>	<i>%</i>
	<i>Number of Fixations</i>				<i>Number of Fixations</i>		
CMU	84.0	87.60 ± 28.10	100	Arial	58.0	60.59 ± 18.67	100
OpenDys	83.0	88.05 ± 30.35	101	Helvetica	63.0	63.29 ± 17.80	104
Arial	87.5	88.76 ± 30.09	101	OpenDys	60.0	63.95 ± 21.43	106
OpenDys It.	87.0	93.31 ± 35.64	107	Times	63.0	64.14 ± 20.08	106
Times	95.5	98.24 ± 27.56	112	Courier	63.0	65.80 ± 19.80	109
Helvetica	93.0	99.23 ± 40.43	113	Myriad	66.0	69.02 ± 21.29	114
Arial It.	89.0	100.22 ± 40.68	115	OpenDys It.	65.5	69.19 ± 20.32	114
Garamond	96.5	101.02 ± 39.39	114	CMU	68.0	71.88 ± 20.46	119
Courier	93.0	101.33 ± 33.67	116	Arial It.	67.0	72.98 ± 25.42	120
Myriad	93.0	101.37 ± 38.33	116	Times It.	72.0	74.33 ± 26.81	123
Times It.	98.0	101.78 ± 29.22	116	Garamond	73.0	80.75 ± 34.02	133
Verdana	101.0	103.33 ± 34.66	118	Verdana	76.0	80.83 ± 37.21	133

We include the relative percentage for *Number of Fixations* with respect to the smallest average value.

Graph Analysis

- CMU, OpenDys and Arial led to significantly less fixations than three other classes of fonts.
- Arial had the smallest mean for number of fixations. Participants had significantly fewer fixations reading with Arial than with Arial It.
- Verdana had the highest mean for number of fixations.

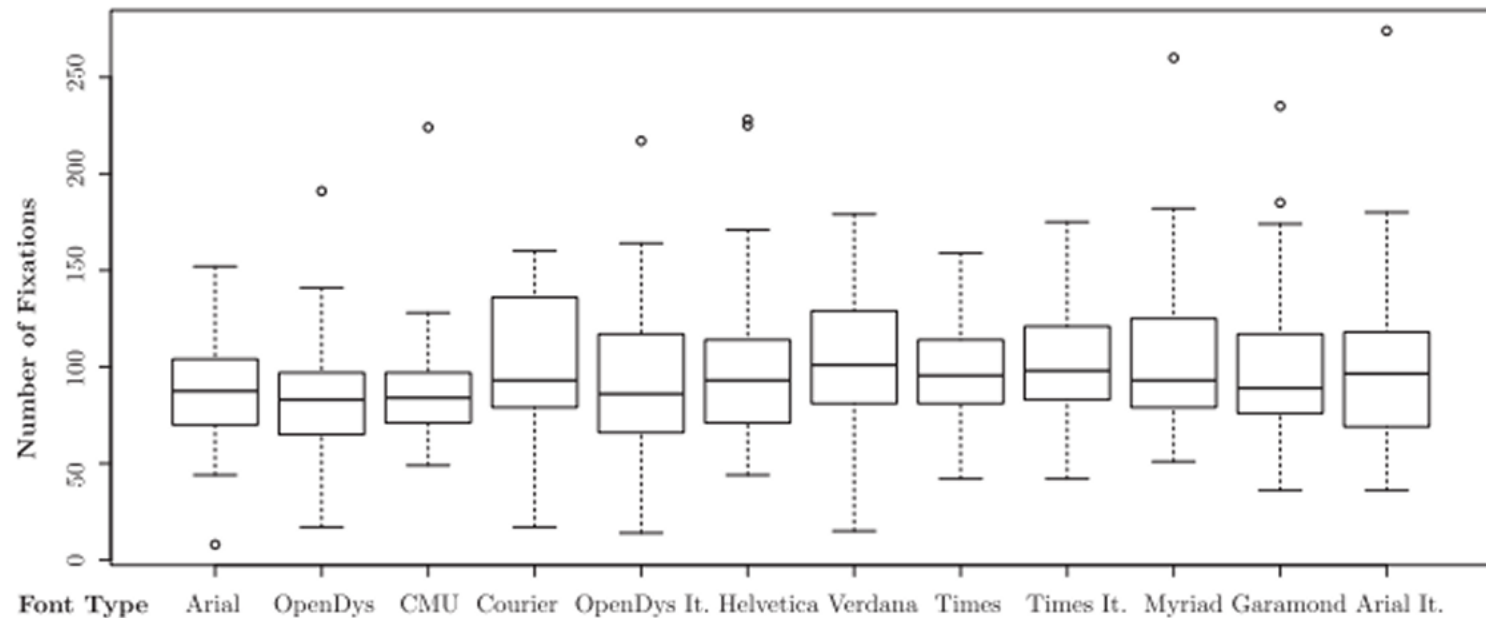
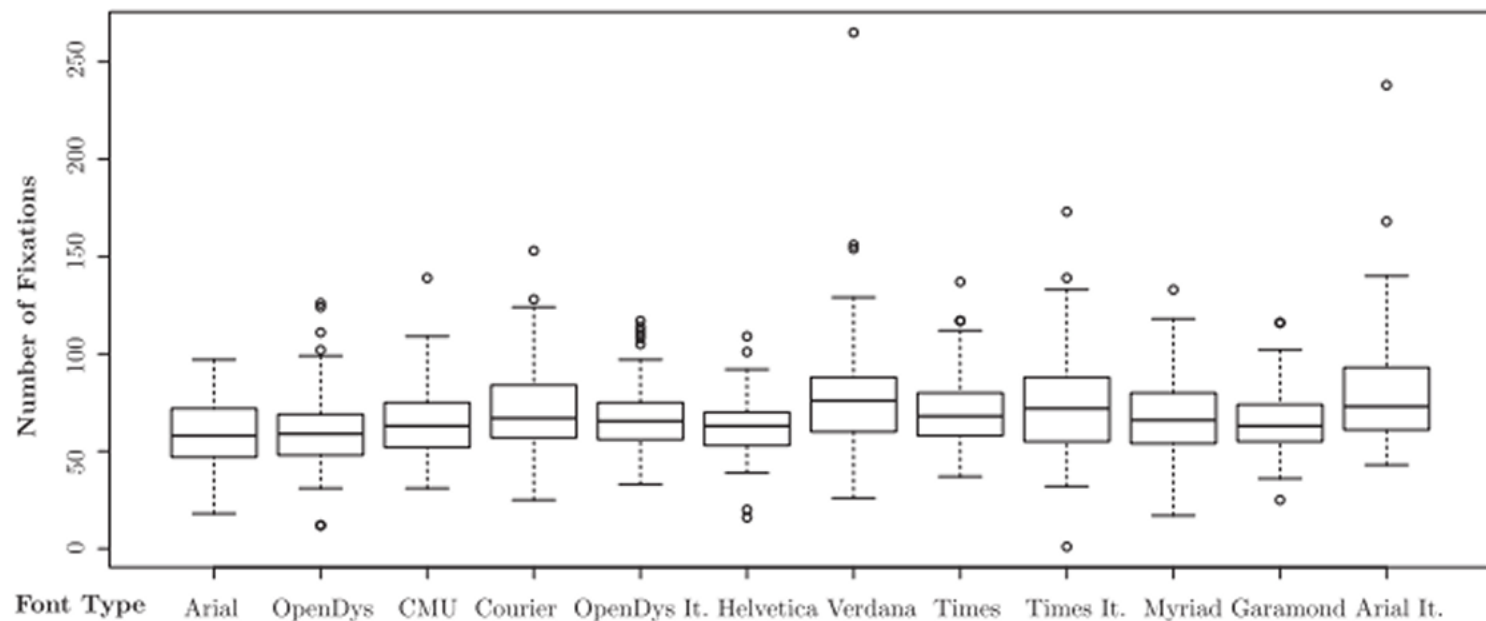


Fig. 8. Number of Fixations box plots by Font Type for group D (ordered by average Reading Time for group D).



<i>Font Type</i>		Group D		<i>Font Type</i>		Group N	
	<i>Median</i>	<i>Mean $\pm$ SD</i>	<i>%</i>		<i>Median</i>	<i>Mean $\pm$ SD</i>	<i>%</i>
	<i>Preference Rating</i>				<i>Preference Rating</i>		
Verdana	4	3.79 $\pm$ 0.98	100	Helvetica	4	3.97 $\pm$ 0.92	100
Helvetica	4	3.62 $\pm$ 1.08	96	Verdana	4	3.97 $\pm$ 0.95	100
Arial	4	3.60 $\pm$ 1.13	95	Arial	4	3.84 $\pm$ 0.80	97
Times	4	3.45 $\pm$ 1.15	91	CMU	4	3.79 $\pm$ 0.90	95
Myriad	3.5	3.40 $\pm$ 0.99	90	Myriad	4	3.66 $\pm$ 0.93	92
CMU	3	3.31 $\pm$ 0.98	87	Times	4	3.64 $\pm$ 0.89	92
Courier	3	3.14 $\pm$ 1.39	83	Arial It.	3	3.36 $\pm$ 1.30	85
Arial It.	3	2.90 $\pm$ 1.10	77	Garamond	3	3.33 $\pm$ 0.93	84
Times It.	3	2.86 $\pm$ 1.20	75	Times It.	3	3.18 $\pm$ 1.07	80
OpenDys	3	2.57 $\pm$ 1.15	68	Courier	3	2.85 $\pm$ 1.08	72
Garamond	2	2.57 $\pm$ 1.15	68	OpenDys	2	2.24 $\pm$ 1.09	56
OpenDys It.	2	2.43 $\pm$ 1.17	64	OpenDys It.	2	2.03 $\pm$ 1.04	51

We include the relative percentage for *Preference Rating* with respect to the largest average value, Verdana.

Reading Time and Preference Rating had a strong negative correlation.

The Aesthetic Trap: subjective vs objective

The Disconnect:

The hidden trap is that users might report liking a familiar font, but eye-tracking says otherwise . Open Dyslexic scored the lowest or one of the lowest in subjective preference however, did very well when it came to reading speed.

However, fonts such as Verdana that scored high on user rating didnt do very well when it came to number of fixations. Thus this tells us users subjective views dont always line up with the effectiveness of a font.

The Accessible Digital Typography Toolkit

Arial

- ✓ Champion of minimal fixations. Best for continuous, uninterrupted reading flow.

Courier

- ✓ Champion of rapid fixation duration. The monospaced uniformity provides maximum predictability for dyslexic readers.

CMU

- ✓ High objective readability; excellent for technical or scientific digital publishing.

Helvetica

- ✓ Exceptional subjective preference paired with ultra-short processing times. The ultimate all-rounder.

Verdana

- ✓ The undisputed king of the screen penalty. Custom-engineered for pixels, it remains the gold standard for subjective comfort and objective retention, especially when double-spaced.

Aesthetics ≠ Accessibility

Designing typography for the margins—accounting for the high friction of digital screens and the biomechanics of dyslexia—does not compromise design. It creates a frictionless, universal environment where the mainstream thrives.

References-

The Effect of Font Type on Screen Readability by People with Dyslexia LUZ RELLO and RICARDO BAEZA-YATES, Web Research Group, DTIC, Universitat Pompeu Fabra, Barcelona, Spain